

Unit 3 - Lines and Angles

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The following B.E.S.T Standards will be covered in this unit:

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

- *Clarification 1: Postulates, relationships and theorems include vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.*
- *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.*
- *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

- *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

MA.912.LT.4.3 Identify and accurately interpret "if...then," "if and only if," "all" and "not" statements. Find the converse, inverse and contrapositive of a statement.

- *Clarification 1: Instruction focuses on recognizing the relationships between an "if...then" statement and the converse, inverse and contrapositive of that statement.*
- *Clarification 2: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

Unit 3, Lesson 1: Proving Angle Relations



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can solve problems that use the relationships of congruent supplements, congruent complements, and vertical angles.
- ☐ I can use the relationships of congruent supplements, congruent complements, and vertical angles in proofs.



3.1.1 Warm-Up: Which One Doesn't Belong?

Four equations are shown.

$$y = 2x$$

$$y = x + 9$$

$$y = -4x + 2$$

$$y = 3x - 2$$

1. Of the four equations, choose one that does not belong.
2. Explain your reasoning for choosing the equation you did.

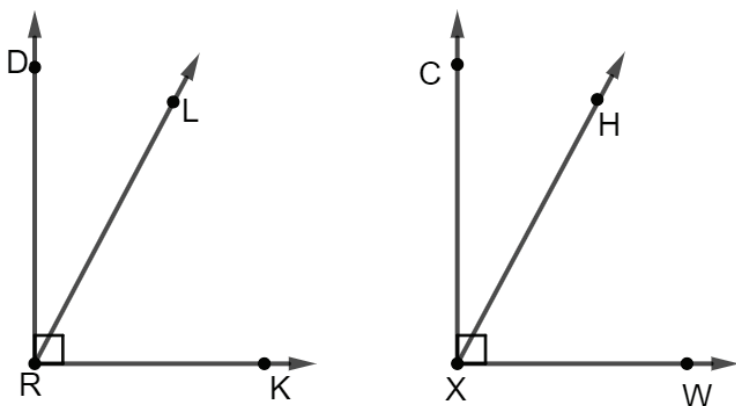


3.1.2 Exploration: Complementary Angles



Complementary angles are two angles whose measures sum to 90° .

1. Right angles $\angle DRK$ and $\angle CXW$ are shown. $\overrightarrow{RL}$ is inside $\angle DRK$ and $\overrightarrow{XH}$ is inside $\angle CXW$. $m\angle DRL = 28^\circ$ and $m\angle CXH = 28^\circ$.

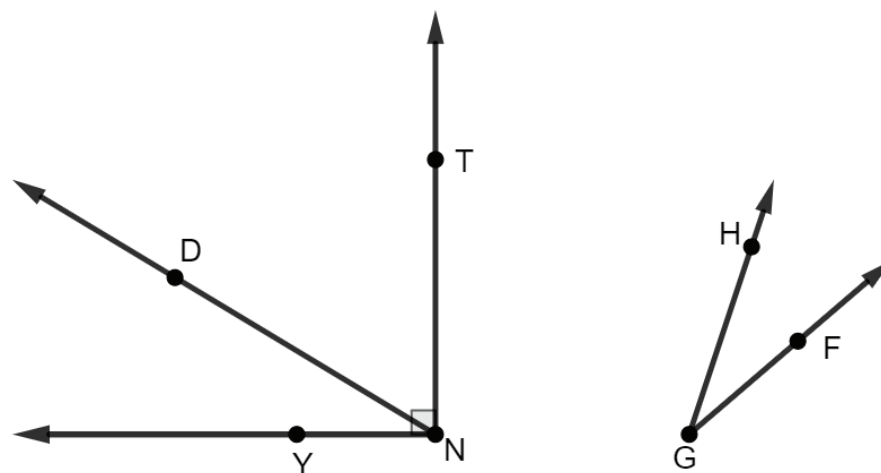


- a. Use the definition of complementary angles to explain why $\angle LRD$ and $\angle LRK$ are complementary angles.

- b. Determine the measure of $\angle LRK$.

- c. Discuss with your partner why $m\angle HXW$ must be the same as $m\angle LRK$. Summarize your discussion.

2. $\angle YNT$ and $\angle FGH$ are shown. $\overrightarrow{ND}$ is inside $\angle YNT$, such that $m\angle YND = 31^\circ$.



- a. What relationship do $\angle YND$ and $\angle DNT$ have?

b. Complete the paragraph proof.

$\angle YNT$ is a(n) ☐ acute
☐ right
☐ obtuse angle. Therefore, by

definition, $m\angle YNT = \underline{\hspace{2cm}}^\circ$. By the angle addition postulate, $m\angle YND + m\angle DNT = m\angle \underline{\hspace{2cm}}$.

Therefore, by the substitution property of equality,

$m\angle YND + m\angle DNT = \underline{\hspace{2cm}}^\circ$.

By the definition of ☐ supplementary
☐ complementary angles,

$\angle YND$ and $\angle DNT$ are ☐ supplementary
☐ complementary angles.

c. On a piece of patty paper, trace $\angle YND$. Determine which rigid transformations can be used to map $\angle YND$ onto $\angle FGH$.

d. Use patty paper or a compass to create $\angle FGW$ so that $m\angle FGW = m\angle YNT$ and $\overrightarrow{GH}$ is inside of $\angle FGW$.

e. Complete the paragraph proof.

Since $m\angle FGW = m\angle YNT$ and $m\angle YNT = \underline{\hspace{2cm}}$, then $m\angle FGW = \underline{\hspace{2cm}}$ by the transitive property of

equality. By the angle addition postulate, $m\angle FGH + m\angle WGH = m\angle \underline{\hspace{2cm}}$. Therefore, by the transitive property of equality,

$m\angle \underline{\hspace{2cm}} + m\angle \underline{\hspace{2cm}} = 90^\circ$ and are

☐ supplementary
☐ complementary angles by the definition of

☐ supplementary
☐ complementary angles.

Since $m\angle YND + m\angle DNT = 90^\circ$, then by the transitive property of equality,

$m\angle \underline{\hspace{2cm}} + m\angle \underline{\hspace{2cm}} = m\angle \underline{\hspace{2cm}} + m\angle \underline{\hspace{2cm}}$. Since $m\angle YND = m\angle FGH$, then by the substitution property $m\angle FGH + m\angle WGH = m\angle \underline{\hspace{2cm}} + m\angle DNT$. Therefore, $m\angle \underline{\hspace{2cm}} = m\angle \underline{\hspace{2cm}}$ by the subtraction property of equality.



Congruent Complements Theorem

Complements of congruent angles are congruent.

3. Use the congruent complements theorem to complete the statements.

$\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle MCY$.

$\angle THM$ is congruent to $\angle WBK$, so $\angle$ _____ is congruent to $\angle$ _____.

A **corollary** is a true statement that is a simple deduction from a theorem or postulate. Its proof requires only a few simple statements in addition to the proof of the original theorem or postulate.



Corollary to Congruent Complements Theorem

Complements of the same angle are congruent.

4. Use the corollary to congruent complements theorem to complete the statement.

$\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle LGX$, so $\angle$ _____ is congruent to $\angle$ _____.

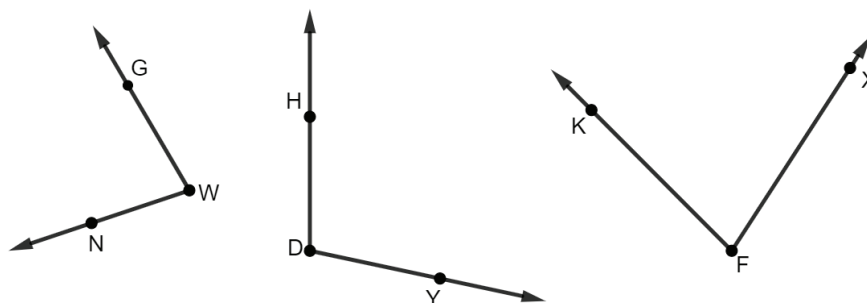


3.1.3 Exploration: Supplementary Angles



Supplementary angles are two angles whose measures sum is exactly 180° .

1. $\angle GWN$, $\angle HDY$, and $\angle KFX$ are shown. $\angle GWN$ is supplementary to $\angle HDY$. $\angle KFX$ is supplementary to $\angle HDY$.



- Determine what relationship, if any, $\angle GWN$ and $\angle KFX$ have.
- Discuss with your partner what can be concluded about $\angle GWN$ and $\angle KFX$. Summarize your discussion.

- c. Ask two classmates their conclusion about $\angle GWN$ and $\angle KFX$. Record their conclusion and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials

- d. Discuss your classmates' conclusions with your partner.



Congruent Supplements Theorem

Supplements of congruent angles are congruent.

2. Use the congruent supplements theorem to complete the statements.

$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle MCY$. $\angle LXG$ is congruent to $\angle BKW$, so $\angle$ _____ is congruent to $\angle$ _____.



Corollary to Congruent Supplements Theorem

Supplements of the same angle are congruent.

3. Use the corollary to congruent supplements theorem to complete the statement.

$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle LXG$, so $\angle$ _____ is congruent to $\angle$ _____.



3.1.4 Guided Instruction: Angles

Two adjacent angles that form a straight angle are called a linear pair.

A **linear pair** is two adjacent angles formed by two intersecting lines.



Linear Pair Postulate

If two angles form a linear pair, then they are supplementary.

1. Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank

Definition of Linear Pair

Linear Pair Postulate

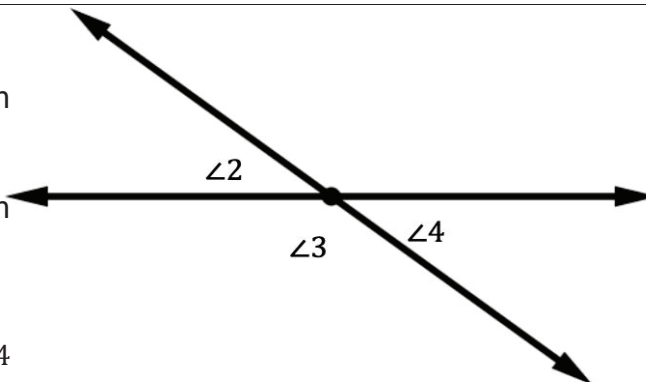
Supplements of the same angle are congruent.

Complements of the same angle are congruent.

Given:

$\angle 2$ and $\angle 3$ form a linear pair.

$\angle 3$ and $\angle 4$ form a linear pair.



Prove: $\angle 2 \cong \angle 4$

Statement	Reason
1. $\angle 2$ and $\angle 3$ form a linear pair	1. Given
2. $\angle 3$ and $\angle 4$ form a linear pair	2. Given
3. $\angle 2$ and $\angle 3$ are supplementary angles	3.
4. $\angle 3$ and $\angle 4$ are supplementary angles	4.
5. $\angle 2 \cong \angle 4$	5.



Vertical angles are the opposite angles formed when two lines intersect.

2. $\angle 2$ and $\angle 4$ are also known as _____ angles.



Vertical Angles Theorem

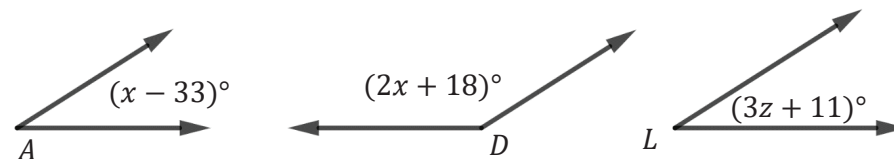
If two lines intersect, then vertical angles are congruent.



3.1.5 Try It: Solving Problems Using Angle Relationships

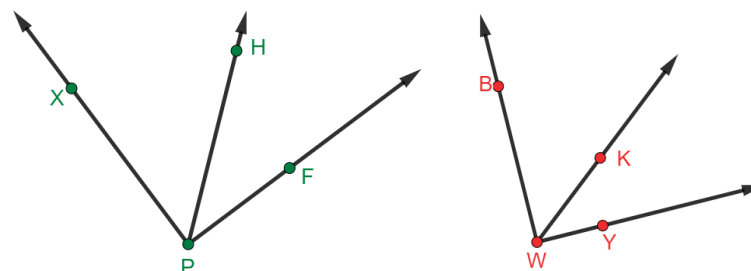
1. Three angles are shown with the given information.

$\angle A$ is supplementary to $\angle D$
 $\angle L$ is supplementary to $\angle D$



What is the value of z ?

2. Right angles $\angle XPF$ and $\angle BWY$ are shown. $\overrightarrow{PH}$ is inside $\angle XPF$, such that $m\angle HPF = 39^\circ$. $m\angle HPF = m\angle KWY$.



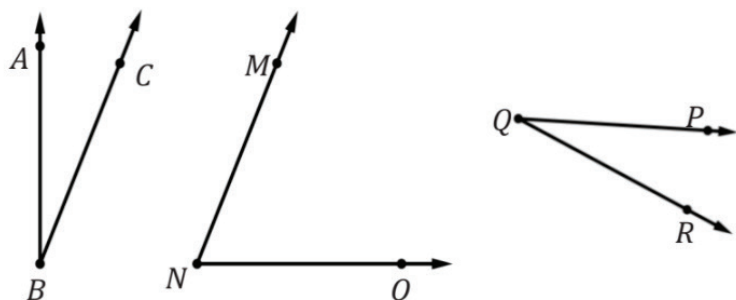
What is $m\angle BWK$?



3.1.6 Wrap-Up: Cool Down

1. Three angles are shown with the given information.

$\angle ABC$ is complementary to $\angle MNO$
 $\angle MNO$ is complementary to $\angle PQR$
 $m\angle ABC = (5x - 3)^\circ$
 $m\angle MNO = 2(x + 22)^\circ$.



- a. Find the value of x .

- b. What is $m\angle PQR$?

Unit 3, Lesson 2: Vertical and Corresponding Angles



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



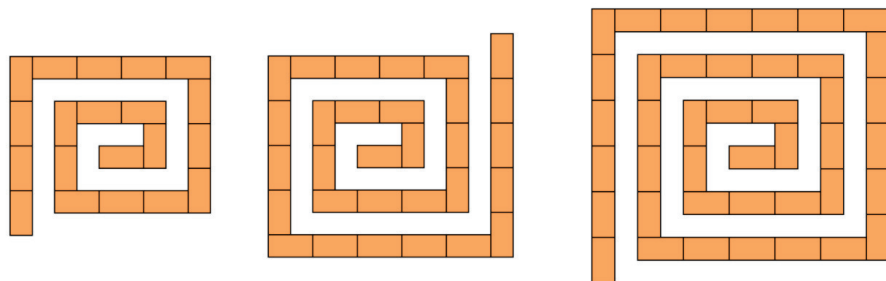
Learning Targets

- ☐ I can use vertical angles to solve problems.
- ☐ I can use corresponding angles formed by parallel lines and a transversal to solve problems.



3.2.1 Warm-Up: Shape Sequence

A sequence of three shapes created using rectangles is shown.

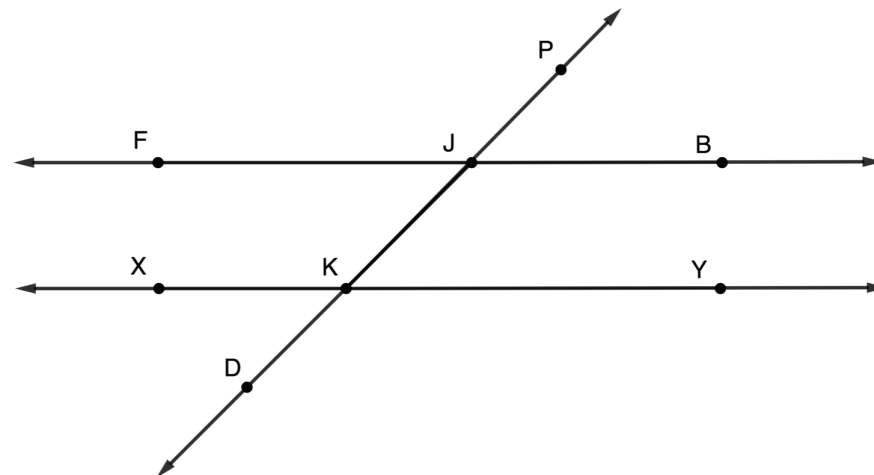


1. How many rectangles will have to be added for the next step in the sequence?
2. How many rectangles will be in Figure 11?



3.2.2 Exploration: Parallel Lines

$\overleftrightarrow{FB}$, $\overleftrightarrow{XY}$, and $\overleftrightarrow{DP}$ are shown, where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{FB}$ at point J and where $\overleftrightarrow{DP}$ intersects $\overleftrightarrow{XY}$ at point K .



1. Discuss with your partner what rigid transformations can be used to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K .
2. How can you use a translation to map $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K ?

In the diagram, $\overleftrightarrow{DP}$ is also known as a transversal, and $\overleftrightarrow{FB}$ and $\overleftrightarrow{XY}$ are parallel lines denoted as $\overleftrightarrow{FB} \parallel \overleftrightarrow{XY}$.



A **transversal** is a line that intersects two or more lines in the same plane at different points.

3. Use the translation of $\overleftrightarrow{FB}$ onto $\overleftrightarrow{XY}$ so that point J coincides with point K to name the new angles in the chart.

Preimage	Image
$\angle PJF$	
	$\angle JKY$
$\angle DJB$	
	$\angle XKD$

The angles listed in the table above are corresponding angles.



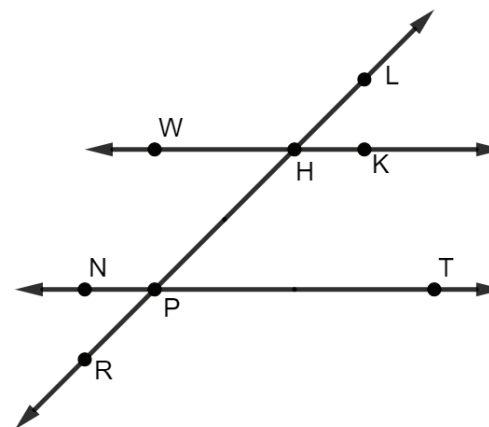
Corresponding angles are angles that are in the same position on two parallel lines in relation to a transversal.

4. Complete the statement.

Given that corresponding angles of parallel lines can be mapped to each other using rigid transformations, it can be concluded that the angles are

- ☐ complementary.
- ☐ congruent.
- ☐ supplementary.

5. In the figure shown, $\overleftrightarrow{WK} \parallel \overleftrightarrow{NT}$ where $\overleftrightarrow{PH}$ is a transversal.



- a. For each angle, name a corresponding angle and a vertical angle.

	$\angle TPR$	$\angle WHP$	$\angle KHP$	$\angle NPR$
Corresponding				
Vertical				

- b. Discuss your answers with your partner and make changes, if necessary.



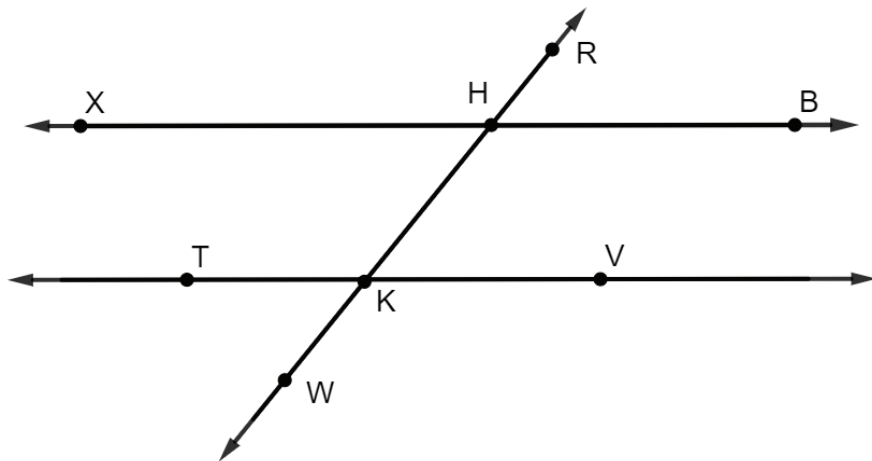
3.2.3 Guided Practice: Angles in Parallel Lines



Corresponding Angles Theorem

If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown such that $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ and $\overleftrightarrow{RW}$ is a transversal.



- a. Given $m\angle RHB = 51^\circ$, determine the measurement of each angle.

$$m\angle HKV =$$

$$m\angle TKW =$$

$$m\angle XHK =$$

- b. Describe how to use $m\angle RHB = 51^\circ$ to find $m\angle XHR$.
- c. Explain why knowing $m\angle XHR$ allows all of the angles to be determined.

2. $\overleftrightarrow{MN}$, $\overleftrightarrow{DK}$, and $\overleftrightarrow{PW}$ lie on a plane such that $\overleftrightarrow{MN} \parallel \overleftrightarrow{DK}$ and $\overleftrightarrow{PW}$ is a transversal. $\overleftrightarrow{PW}$ intersects $\overleftrightarrow{MN}$ at point Y and $\overleftrightarrow{DK}$ at point F . $m\angle PYN = (4x + 36)^\circ$, $m\angle NYF = (11y)^\circ$, and $m\angle KFW = 44^\circ$.

a. Draw a diagram that represents the given information.

b. Determine the value of y .

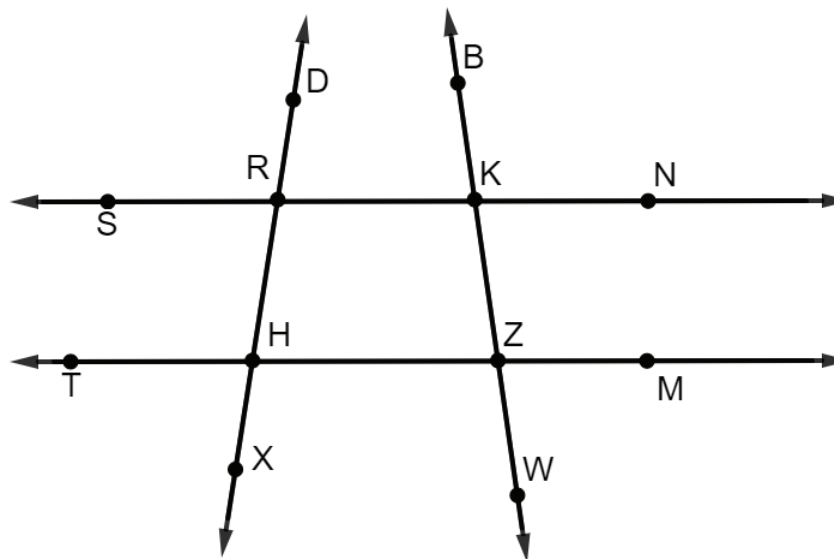
c. Determine the value of x .

d. What is $m\angle PYM$?



3.2.4 Your Turn!

1. $\overleftrightarrow{SN}$, $\overleftrightarrow{TM}$, $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ lie on a plane, such that $\overleftrightarrow{SN} \parallel \overleftrightarrow{TM}$ and $\overleftrightarrow{DX}$ and $\overleftrightarrow{BW}$ are transversals. $m\angle BKN = 99^\circ$ and $m\angle KRH = 99^\circ$.

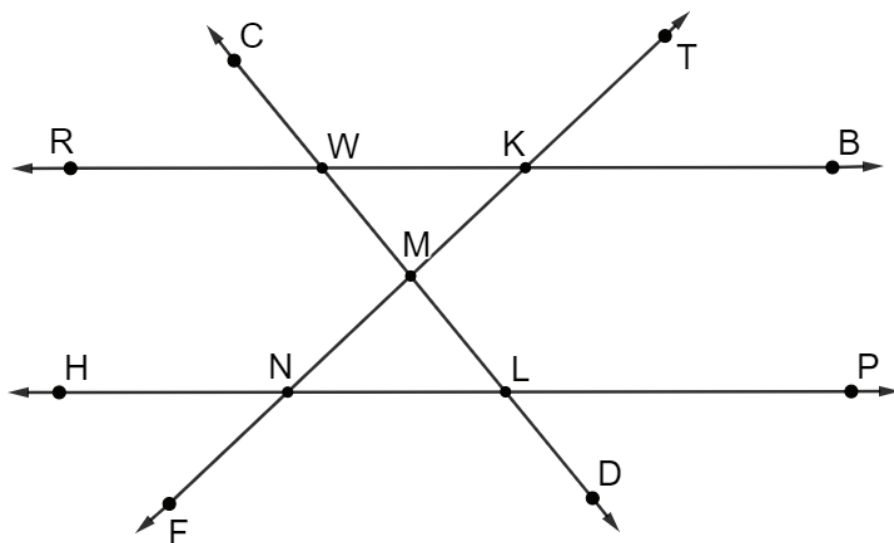


a. What is $m\angle KZM$?

b. What is $m\angle ZHX$?

c. What is $m\angle DRS$?

2. $\overleftrightarrow{CD}$, $\overleftrightarrow{TF}$, $\overleftrightarrow{RB}$, and $\overleftrightarrow{HP}$ are shown where $\overleftrightarrow{RB} \parallel \overleftrightarrow{HP}$ and $m\angle CWR = 51^\circ$ and $m\angle TKB = 43^\circ$.



Jasmyn's and Drew's justifications of the measure of $\angle CLH$ is shown.

Jasmyn's Answer	Drew's Answer
$m\angle CLH = 51^\circ$ because $\angle CLH$ and $\angle CWR$ are corresponding angles.	$m\angle CLH = 43^\circ$ because $\angle TKB$ and $\angle CWR$ are corresponding angles.

- a. Explain which student you agree with, including your reasoning.

- b. Find the measure of the following angles.

$$m\angle KNL = \underline{\hspace{2cm}}$$

$$m\angle HNF = \underline{\hspace{2cm}}$$

$$m\angle MLN = \underline{\hspace{2cm}}$$



3.2.5 Wrap-Up: Cool Down

1. Draw an example of vertical angles.
2. Draw an example of corresponding angles.

Unit 3, Lesson 3: Congruent Angles in Parallel Lines



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



Learning Target

- ☐ I can use postulates and theorems to find unknown angle measures when parallel lines are intersected by a transversal.



3.3.1 Warm-Up: Using Math in Making Guitars

Patrick Evans is a production manager at Maton Guitars in Australia. Country music star Keith Urban uses Maton guitars. The precision necessary in making each guitar requires mathematics, or “maths,” as Australians call it.



1. What was one thing you found interesting about the process of making a guitar?



3.3.2 Guided Instruction: Angle Relationships

1. In each diagram, $\overline{TM} \parallel \overline{NW}$ where $\overline{YL}$ is a transversal.
 - a. Use patty paper to determine if the angles marked using color are congruent.

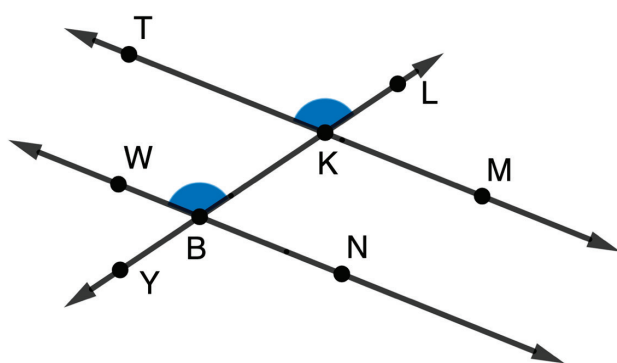
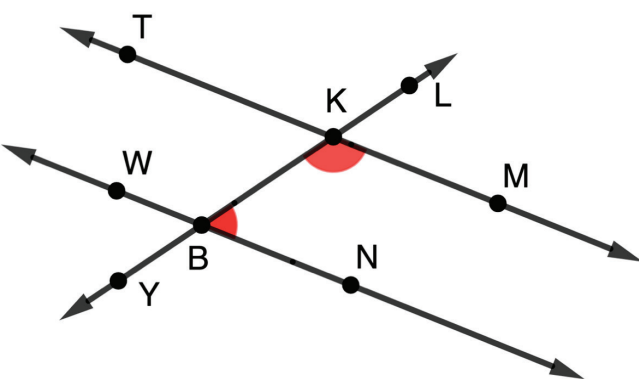
Diagram	Congruent?
A 	☐ Yes ☐ No
B 	☐ Yes ☐ No

Diagram	Congruent?
C	☐ Yes ☐ No
D	☐ Yes ☐ No



Alternate Exterior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.

b. Which diagram shows a pair of angles that are alternate exterior angles?

- ☐ Diagram A
- ☐ Diagram B
- ☐ Diagram C
- ☐ Diagram D



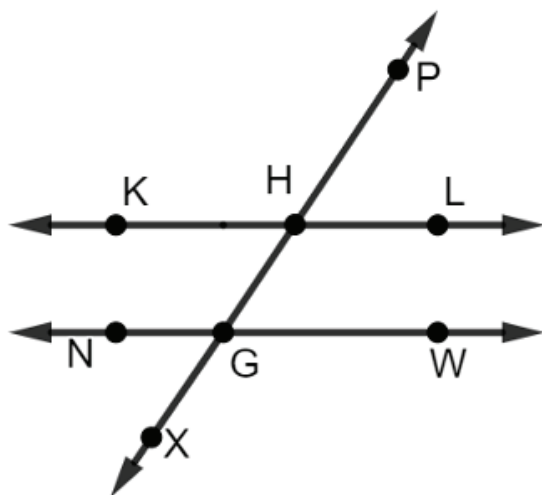
Alternate Interior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.

c. Which diagram shows a pair of angles that are alternate interior angles?

- ☐ Diagram A
- ☐ Diagram B
- ☐ Diagram C
- ☐ Diagram D

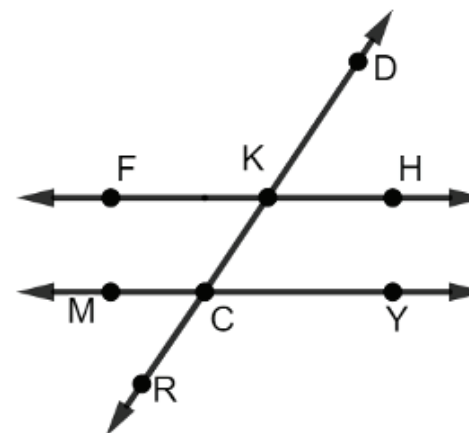
2. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$, where $\overleftrightarrow{XP}$ is a transversal.



Work with your partner to name each pair of angles as corresponding, alternate exterior, alternate interior angles, or none of these.

Angles	Type
$\angle WGX$ and $\angle KHP$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle LHG$ and $\angle WGX$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle NGX$ and $\angle KHP$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle KHG$ and $\angle WGH$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above

3. Given: $\overleftrightarrow{FH} \parallel \overleftrightarrow{MY}$
 Prove: $\angle DKH \cong \angle RCM$



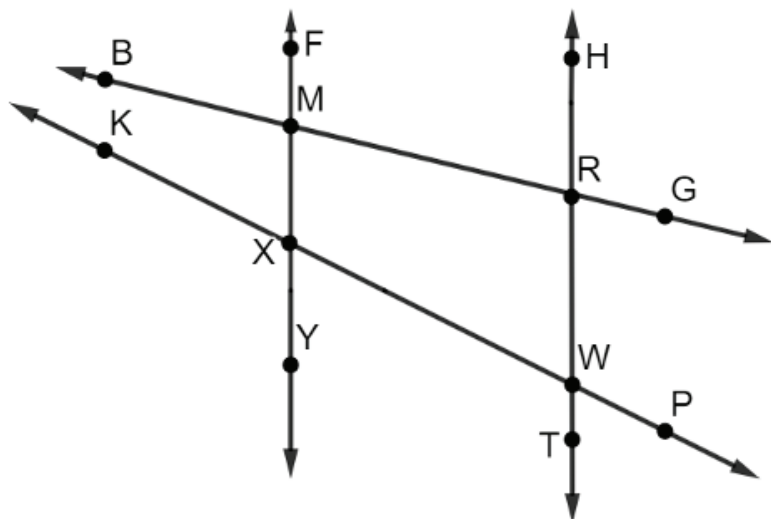
Fill in the table to complete the proof.

Statement	Reason
1.	1. Given
2.	2. Vertical angles are congruent.
3.	3.
4.	4. Transitive Property of Congruence



3.3.3 Try It: Angles and Parallel Lines

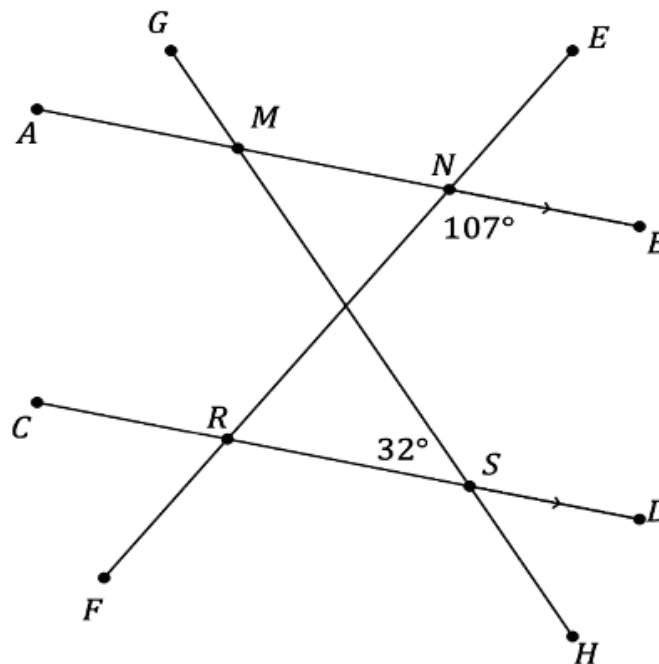
1. $\overleftrightarrow{FY} \parallel \overleftrightarrow{HT}$ where $\overleftrightarrow{BG}$ and $\overleftrightarrow{KP}$ are transversals and $m\angle BMF = 76^\circ$ and $m\angle KXY = 117^\circ$.



Complete the table.

Angle	Angle Measure
$\angle MRW$	
$\angle PWT$	
$\angle YXW$	
$\angle RMX$	
$\angle PWR$	

2. $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ where $\overleftrightarrow{GH}$ and $\overleftrightarrow{EF}$ are transversals and $m\angle BNR = 107^\circ$ and $m\angle RSM = 32^\circ$.



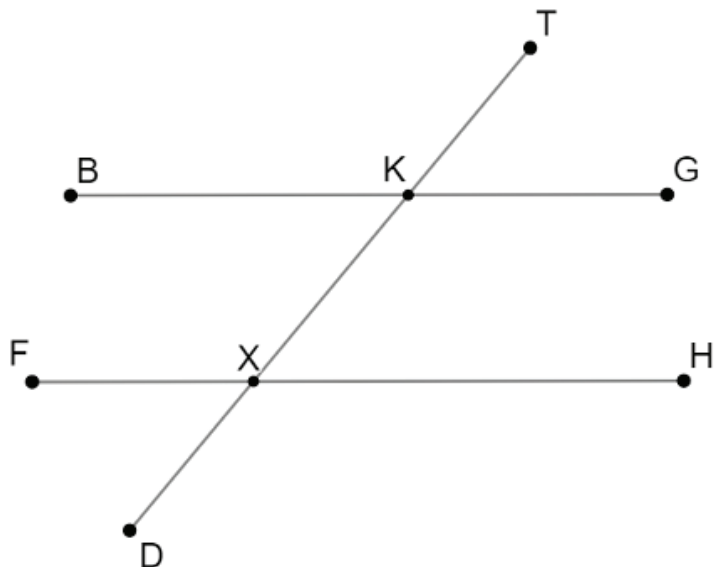
Complete the table.

Angle Measure	Postulate or Theorem Used
$m\angle NRC =$	
$m\angle BMH =$	
$m\angle DRF =$	
$m\angle GMA =$	



3.3.4 Your Turn!

1. $\overline{BG} \parallel \overline{FH}$ where $\overline{TD}$ is a transversal and $m\angle HXD = 134^\circ$.



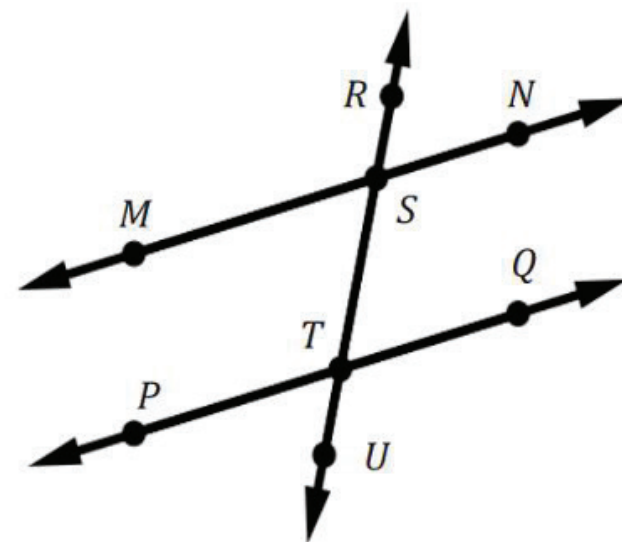
Find the measure of each angle. Then, state the angle relationships used to determine the value.

Angle Measure	Angle Relationships
$m\angle GKK =$	
$m\angle FKK =$	
$m\angle BKT =$	



3.3.5 Wrap-Up: Ticket Out the Door

1. $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$ and $m\angle PTS = 100^\circ$.



- In relation to the parallel lines, what angle relationship do $\angle TSN$ and $\angle PTS$ have?
- What is $m\angle TSN$?

Unit 3, Lesson 4: Consecutive Angles



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



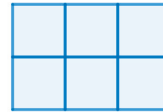
Learning Targets

- ☐ I can determine the relationships of consecutive exterior and interior angles.
- ☐ I can solve problems that involve angles formed by parallel lines intersected by a transversal.

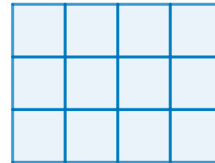


3.4.1 Warm-Up: Squares in Rectangles

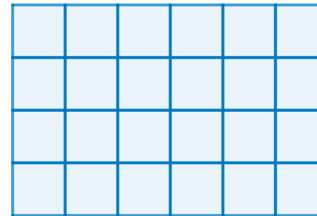
Three rectangles and their descriptions are shown.



A 2×3 rectangle contains 6 unit squares.



A 3×4 rectangle contains 12 unit squares.



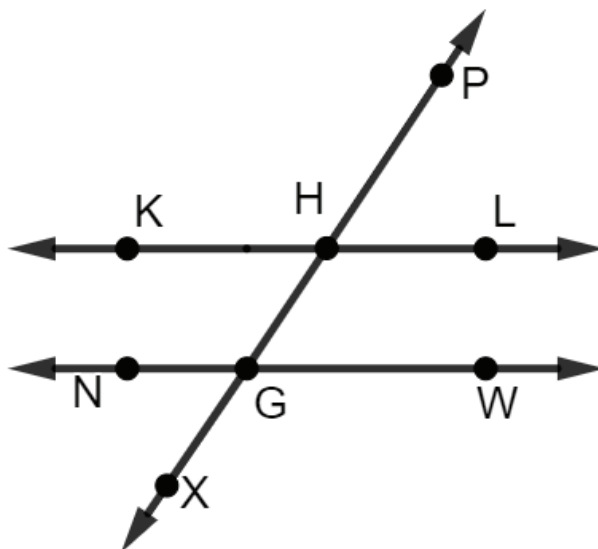
A 4×6 rectangle contains 24 unit squares.

1. What are the dimensions of a rectangle that contains exactly 100 unit squares?
2. List any other size rectangles that contain exactly 100 unit squares.



3.4.2 Collaboration: Consecutive Interior and Exterior Angles

1. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$ where $\overleftrightarrow{XP}$ is a transversal.



- a. Work with your partner to determine if each pair of angles could be congruent by using corresponding, alternate exterior, alternate interior, or vertical angles.

Angles	Congruent?
$\angle NGH$ and $\angle KHG$	
$\angle PHL$ and $\angle WGX$	

- b. Work with your partner to determine which pair of angles in the table could be described as consecutive interior angles.

- c. Name another pair of consecutive interior angles.
- d. Work with your partner to write a conjecture about the relationship that is true for consecutive interior angles.
- e. Ask another classmate for their conjecture about consecutive interior angles. Record their conjecture and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials

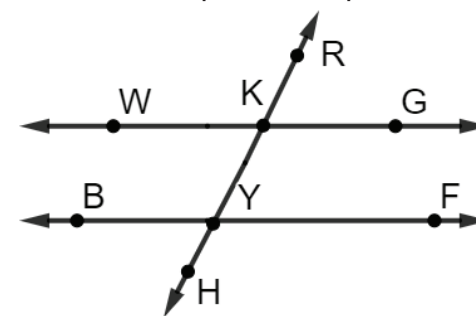
- f. Compare the conjecture from your classmate with the conjecture you and your partner wrote. If necessary, make any adjustments to your conjecture.
- g. $\angle PHL$ and $\angle WGX$ are consecutive exterior angles. Name another pair of consecutive exterior angles.
- h. Work with your partner to write a true conjecture about the relationship between consecutive exterior angles.

2. A partially completed proof is shown.

- a. Work with your partner to complete the proof.

Given: $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$

Prove: $\angle WKY$ and $\angle BYK$ are supplementary.

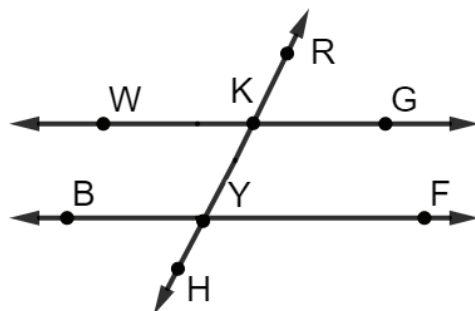


Statement	Reason
1. $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$	1. Given
2. $\angle WKY \cong$ _____	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle BYH$ and $\angle BYK$ are a linear pair	3. Definition of a linear pair
4.	4. If two angles form a linear pair, then they are supplementary.
5. $m\angle BYH + m\angle BYK = 180^\circ$	5. Definition of supplementary angles
6. $m\angle WKY =$ _____	6.
7. $m\angle$ _____ $+ m\angle$ _____ $= 180^\circ$	7. Substitution
8. $\angle WKY$ and $\angle BYK$ are supplementary.	8. Definition of supplementary angles

- b. Check your proof with another classmate. If necessary, make any adjustments.

3. Work with your partner to complete the proof.

Given: $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$
 Prove: $\angle RKG$ and $\angle FYH$
 are supplementary.



Statement	Reason
1. $\overleftrightarrow{WG} \parallel \overleftrightarrow{BF}$	1. Given
2. $\angle RKG \cong$ _____	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle KYF$ and $\angle FYH$ are a linear pair	3. Definition of a linear pair
4.	4. If two angles form a linear pair, then they are supplementary.
5.	5. Definition of supplementary angles
6.	6.
7. $m\angle$ _____ + $m\angle$ _____ = 180°	7. Substitution
8. $\angle RKG$ and $\angle FYH$ are supplementary.	8. Definition of supplementary angles

4. Mistakes can be an opportunity to learn.

- a. What mistakes did you make in the first proof that helped you in the second proof?

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

- b. If you did not make a mistake in the first proof, describe how the proofs were similar.



Consecutive Interior Angles Theorem

If two parallel lines are intersected by a transversal, then interior angles on the same side of the transversal are supplementary.



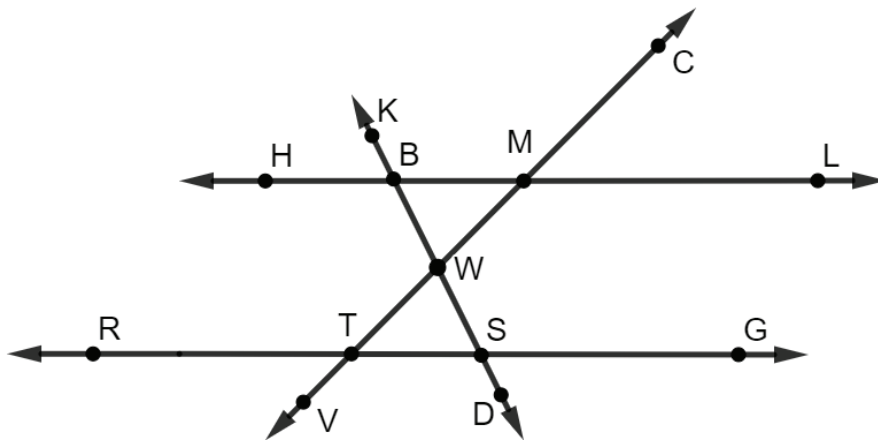
Consecutive Exterior Angles Theorem

If two parallel lines are intersected by a transversal, then exterior angles on the same side of the transversal are supplementary.



3.4.3 Guided Practice: Angles in Parallel Lines

1. $\overleftrightarrow{HL} \parallel \overleftrightarrow{RG}$ where $\overleftrightarrow{CV}$ and $\overleftrightarrow{KD}$ are transversals with $m\angle KBM = 117^\circ$ and $m\angle VTS = 134^\circ$.



Determine the measurement of each angle.

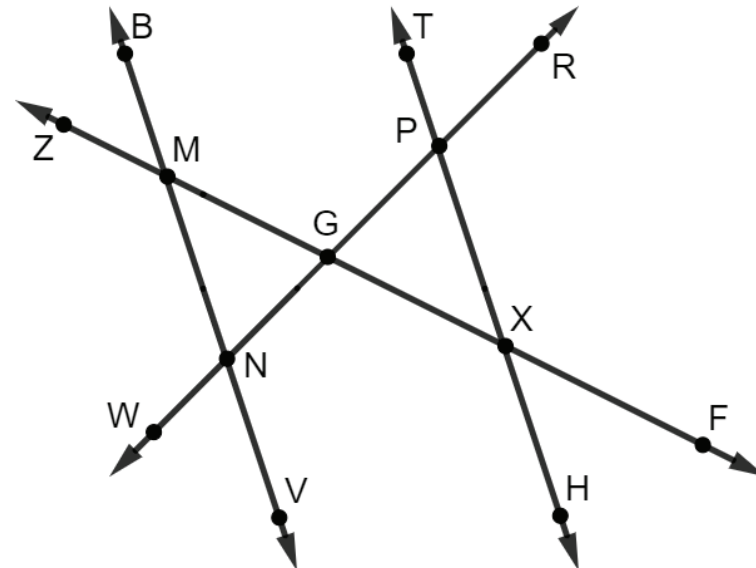
$$m\angle DSG =$$

$$m\angle LMC =$$

$$m\angle RTV =$$

$$m\angle BWM =$$

2. In the figure shown, $\overleftrightarrow{BV} \parallel \overleftrightarrow{TH}$, $m\angle GXH = (5(2x - 1))^\circ$, $m\angle GMN = (4x - 11)^\circ$, $m\angle XPG = (3y - 8)^\circ$, and $m\angle VNG = (7y - 52)^\circ$.



- a. Find the value of x .

- b. Find the value of y .



3.4.4 Wrap-Up: Reflection

1. Rate your confidence in solving problems using these angle pairs.

I can solve number problems involving....	I am an expert and could help others.	I can hold my own.	I am getting there.	I would like more help or practice.
Congruent Supplements				
Congruent Complements				
Vertical Angles				
Consecutive Interior Angles				
Consecutive Exterior Angles				
Alternate Interior Angles				
Alternate Exterior Angles				
Corresponding Angles				



Unit 3, Lesson 5: Converse of the Alternate Interior Angles Theorem



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.



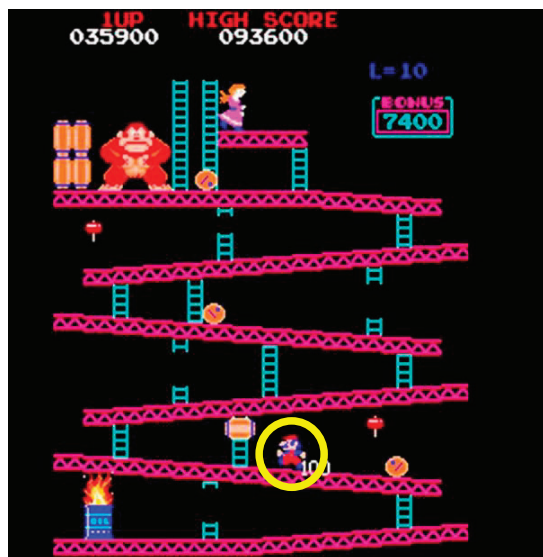
Learning Targets

- ☐ I can write and use the converse of a postulate or theorem.
- ☐ I can use postulates and theorems about parallel lines to determine if lines are parallel.



3.5.1 Warm-Up: Notice and Wonder

The game *Donkey Kong* was released for sale in the U.S. in 1981 by Nintendo, a Japanese playing card and toy company turned video game developer. Mario, circled in yellow, is shown in the image running and climbing up the platforms and ladders.

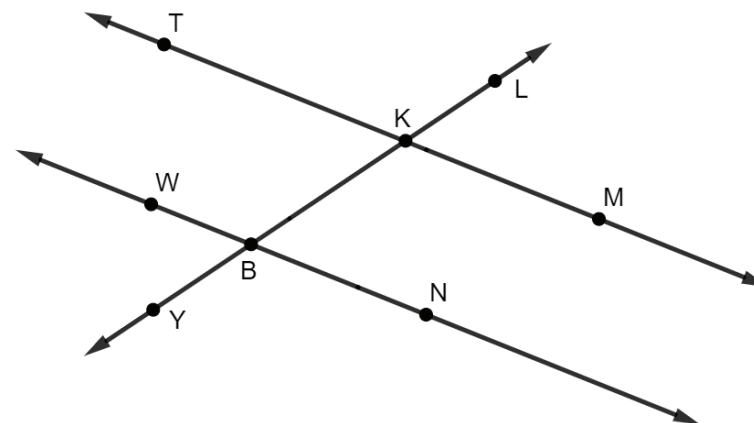


1. What do you notice about the different platforms Mario is running on?
2. What do you wonder?
3. How can you determine if any of the platforms or ladders are parallel?



3.5.2 Exploration: Congruent Angles Formed by Two Lines and a Transversal

1. $\overleftrightarrow{TM}$, $\overleftrightarrow{WN}$, and $\overleftrightarrow{YL}$ are shown. $\overleftrightarrow{YL}$ intersects $\overleftrightarrow{TM}$ and $\overleftrightarrow{WN}$.



Work with your partner to conjecture if the given information in each row of the table is sufficient to show that the lines are parallel.

Angles	Type of Angles	Sufficient?
$\angle TKL \cong \angle WKB$		☐ Yes ☐ No
$\angle MKB \cong \angle WKB$		☐ Yes ☐ No
$\angle MKB \cong \angle TKL$		☐ Yes ☐ No
$\angle TKL \cong \angle YBN$		☐ Yes ☐ No
$\angle WKB$ and $\angle NKB$ are supplementary.		☐ Yes ☐ No
$\angle TKL$ and $\angle WBY$ are supplementary.		☐ Yes ☐ No



3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems

1. Work with your partner or group to write the converse of the postulate or theorem your group is assigned. Then, draw a picture to represent the relationship described.

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.	
Picture	

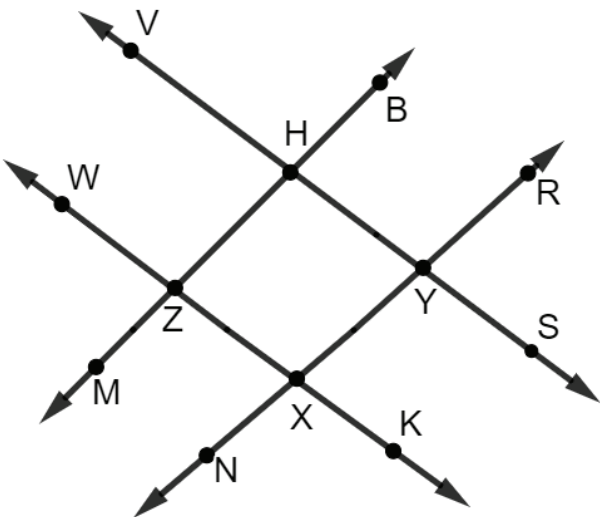
Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then corresponding angles are congruent.	
Picture	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then consecutive exterior angles are supplementary.	
Picture	



3.5.4 Guided Practice: Determining if Lines are Parallel

1. $\overleftrightarrow{VS}$, $\overleftrightarrow{WK}$, $\overleftrightarrow{MB}$, and $\overleftrightarrow{NR}$ are shown where $m\angle VHB = 98^\circ$, $m\angle RYS = 78^\circ$, $m\angle NXK = 102^\circ$, and $m\angle WZM = 82^\circ$.



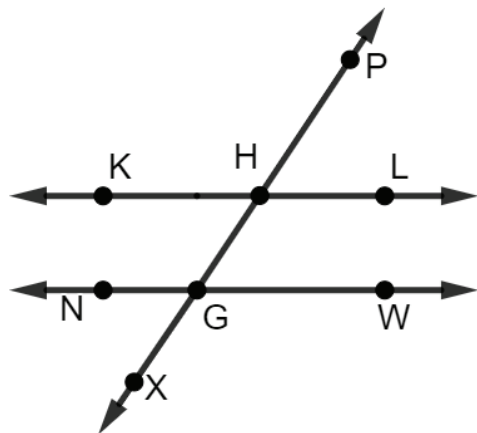
Which pair(s) of lines are parallel? Justify your answer.

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.	
Picture	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then consecutive interior angles are supplementary.	
Picture	



2. $\overleftrightarrow{KL}$, $\overleftrightarrow{NW}$, and $\overleftrightarrow{PX}$ are shown.



Determine if the given condition could be used to justify that $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$?	Justification
$\angle PHL \cong \angle WGX$	☐ Yes ☐ No	
$m\angle KHP + m\angle NGX = 180^\circ$	☐ Yes ☐ No	
$\angle KHG \cong \angle WGH$	☐ Yes ☐ No	
$\angle NGX$ and $\angle WGX$ are supplementary.	☐ Yes ☐ No	
$\angle LHG \cong \angle KHP$	☐ Yes ☐ No	



3.5.5 Wrap-Up: Cool Down

- Write a short note to a friend who missed today's class summarizing what you learned.

Unit 3, Lesson 6: Using Angle Relationships, Postulates, and Theorems



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



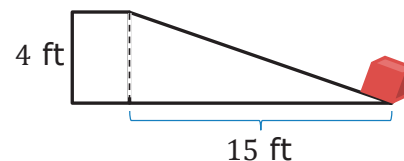
Learning Target

- ☐ I can solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

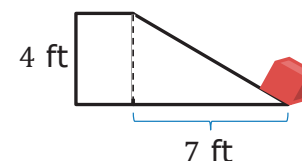


3.6.1 Warm-Up: Would You Rather?

- Two ramps set up to move boxes up to a 4 foot (ft) high platform are shown.



Ramp A



Ramp B

- Would you rather push a 50 pound (lb) box up Ramp A or Ramp B?
- Justify your choice.



3.6.2 Activity: Solving Using Angle Relationships, Postulates, and Theorems

With your partner, sort the cards to match each image with the type of angle pair represented and the value of x . Record your answers in the table, showing your work in the last column.

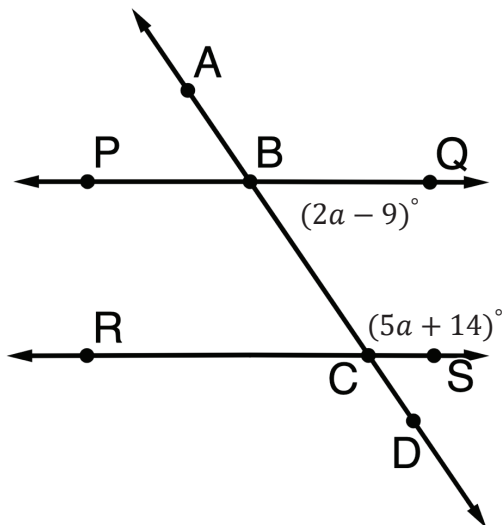
Type of Angle Pair	Letter of Image	Work and Answer

Type of Angle Pair	Letter of Image	Work and Answer



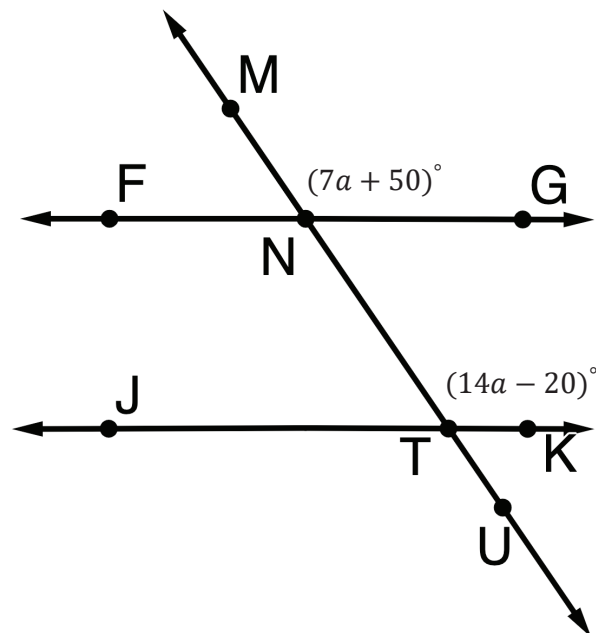
3.6.3 Your Turn!

1. $\overleftrightarrow{PQ} \parallel \overleftrightarrow{RS}$, where $\overleftrightarrow{AD}$ is a transversal.



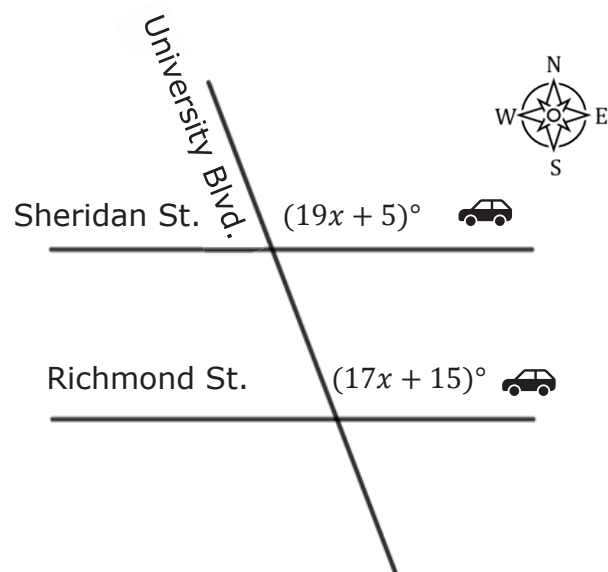
- Determine the value of a .
- What is $m\angle ABQ$?
- What is $m\angle BCR$?

2. $\overleftrightarrow{FG} \parallel \overleftrightarrow{JK}$ where $\overleftrightarrow{MU}$ is a transversal.



- Determine the value of a .
- What is $m\angle FNT$?
- What is $m\angle KTU$?

3. Sheridan Street and Richmond Street are parallel, and both are intersected by University Boulevard.

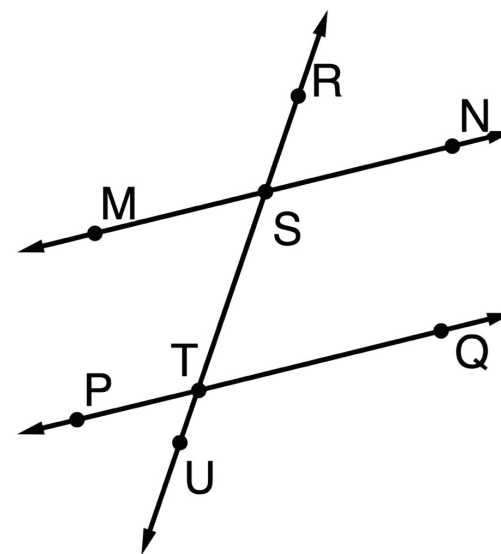


Two cars are heading west, one on Sheridan St. and one on Richmond St. Both cars turn north onto University Blvd. What is the measure of the angle of their turn onto University Blvd.?



3.6.4 Wrap-Up: Ticket Out the Door

1. Given $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$, $m\angle PTS = (14x - 10)^\circ$, and $m\angle NST = (9x + 35)^\circ$, find each angle measure.



a. $m\angle MSR$

b. $m\angle STQ$

Unit 3, Lesson 7: Solving Problems Using Angle Pair Relationships



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



Learning Target

- ☐ I can solve mathematical and real-world problems involving angle pair relationships formed by transversals and parallel lines.



3.7.1 Warm-Up: Angle Pair Relationships

- The image shows two streets intersected by a third street. Various points of interest are located at the intersections.



Match the listed points of interest with their angle relationship listed in the right column. Write the letter of the angle relationship on the line.

Points of Interest	Angle Relationship
_____ Museum and Worship Center	A. Alternate Interior Angles
_____ Bank and Day Care	B. Consecutive Interior Angles
_____ Park and Coffee Shop	C. Corresponding Angles
_____ Fire Station and Worship Center	D. Vertical Angles
_____ Community Center and Day Care	E. Alternate Exterior Angles
_____ Bank and Museum	F. Linear Pair



3.7.2 Collaboration: Solving Problems Using Angle Relationships

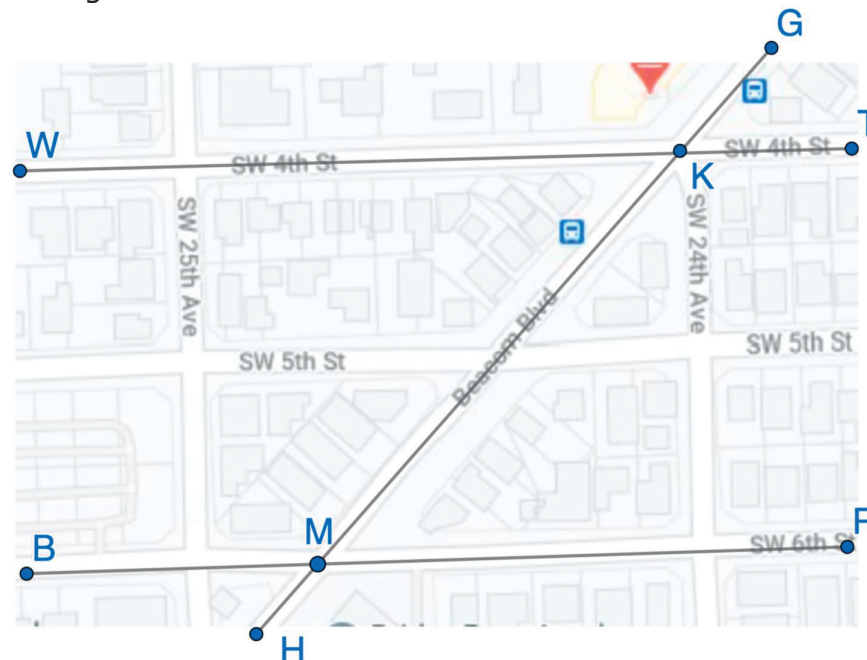
Work with your partner to solve each problem.

1. Mako is a steel roller coaster located at Sea World in Orlando. Some of the support beams are shown in the diagram using $\overline{KL}$, $\overline{LR}$, and $\overline{FR}$.



Determine the value of x that shows that $\overline{KL} \parallel \overline{FR}$, if $m\angle KLR = (7x - 5.6)^\circ$ and $m\angle FRL = (3x + 5.6)^\circ$.

2. Most of the streets in Miami are on a grid. Beacom Blvd. is one of the few streets that does not run north to south or east to west. The diagram shows a map of Beacom Blvd., $\overline{HG}$, between SW 4th Street, $\overline{WT}$, and SW 6th Street, $\overline{BR}$, just a few blocks from Miami Senior High School.



- a. Given that $\overline{WT} \parallel \overline{BR}$, determine the value of x if $m\angle HMB = (2x - 7)^\circ$, $m\angle WKG = (4x + 25)^\circ$, and $m\angle KMB = (6x - 29)^\circ$.

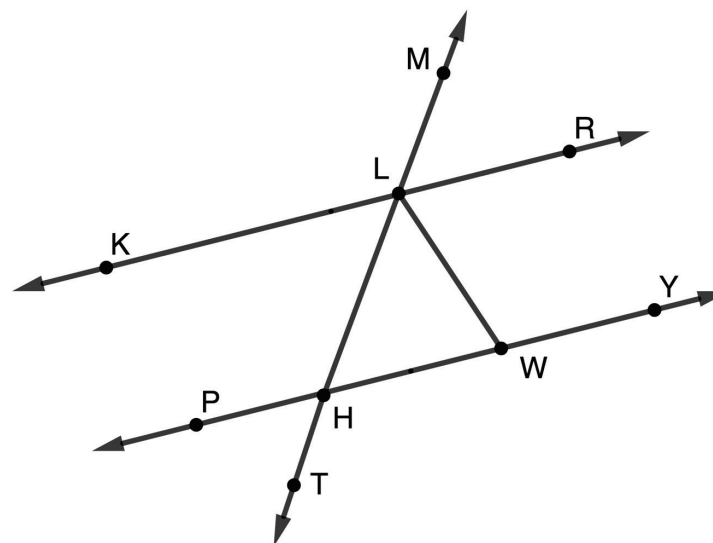
- b. Determine the measure of each angle.

$m\angle HMB$

$m\angle HMR$

$m\angle TKM$

3. $\overleftrightarrow{KR}$, $\overleftrightarrow{PY}$, and $\overleftrightarrow{MT}$ are shown with $\overleftrightarrow{LW}$ where $m\angle LWH = (4y - 9\frac{1}{3})^\circ$, $m\angle WLH = (3y + 3)^\circ$, and $m\angle LHW = m\angle WLH$.



- a. Determine the value of y that shows that $\overleftrightarrow{PY} \parallel \overleftrightarrow{KR}$.

- b. Determine the measure of each angle.

$m\angle RLW$

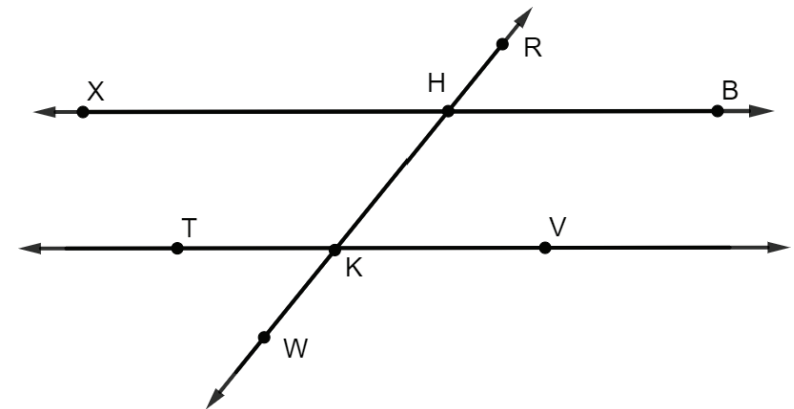
$m\angle PHT$

$m\angle KLM$



3.7.3 Your Turn!

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown. $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ where $\overleftrightarrow{RW}$ is a transversal with $m\angle XHR = (9x + 21)^\circ$ and $m\angle HKV = (4x + 3)^\circ$.



- a. Determine the value of x .
- b. Find the measure of each angle.

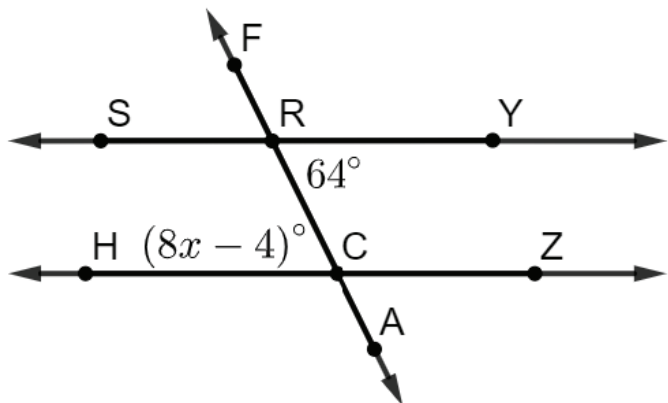
$m\angle XHR$

$m\angle HKV$

$m\angle TKW$

$m\angle BHK$

2. $\overleftrightarrow{SY} \parallel \overleftrightarrow{HZ}$, $m\angle YRC = 64^\circ$, and $m\angle RCH = (8x - 4)^\circ$.



- a. Find the value of x .
- b. Find the measure of each angle.

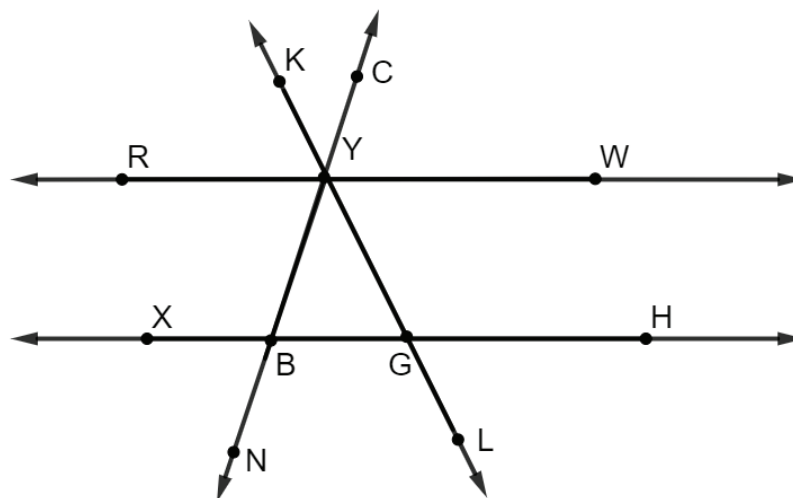
$m\angle ZCA$

$m\angle FRS$

$m\angle YRF$

$m\angle HCA$

3. $\overleftrightarrow{RW} \parallel \overleftrightarrow{XH}$, $m\angle RYK = s^\circ$, $m\angle CYW = m^\circ$, $m\angle GYB = 43^\circ$, $m\angle YBG = 72^\circ$, and $m\angle LGH = (6t + 11)^\circ$.



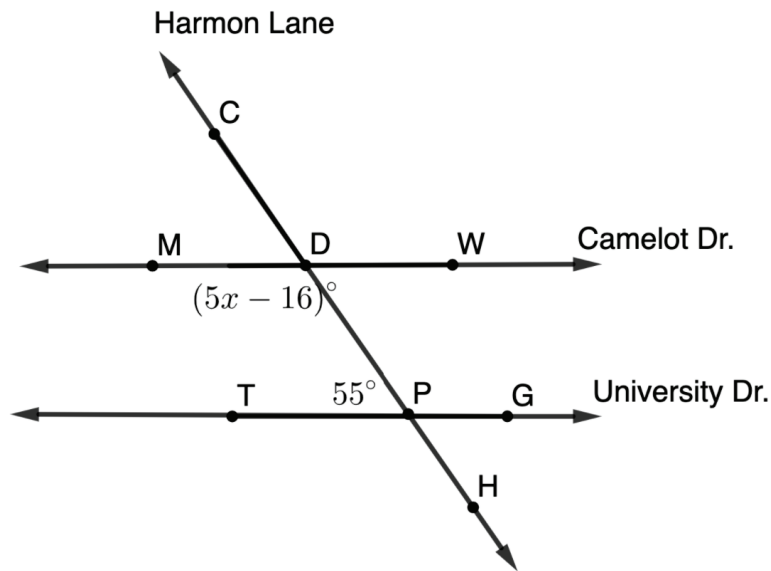
Find the value of each variable. Show your work.

t

m

s

4. Harmon Lane, $\overleftrightarrow{CH}$, crosses the streets Camelot Drive, $\overleftrightarrow{MW}$, and University Drive, $\overleftrightarrow{TG}$, as shown. $m\angle TPD = 55^\circ$, and $m\angle MDP = (5x - 16)^\circ$.



Determine the value of x that shows $\overleftrightarrow{MW} \parallel \overleftrightarrow{TG}$.



3.7.4 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

- I can use corresponding angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



- I can use alternate interior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



- I can use alternate exterior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



- I can use consecutive interior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



- I can use consecutive exterior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



Unit 3, Lesson 8: Angle Relationship Proofs – Part 1



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

- ☐ I can prove relationships using parallel lines and their transversals.



3.8.1 Warm-Up: Would You Rather?

- The meals at two restaurants range from \$8.00 to \$25.00. Servers are paid based on the following.

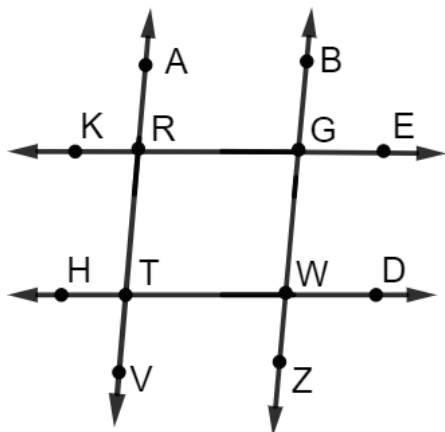
<u>Restaurant A</u>	<u>Restaurant B</u>
\$18 per hour	\$10.50 per hour
No tipping allowed	Tipping encouraged

Would you rather work as a server at Restaurant A or at Restaurant B? Justify your choice.



3.8.2 Collaboration: Parallel Lines

1. Contradictory information is given for the figure.
 $\overleftrightarrow{KE} \parallel \overleftrightarrow{HD}$, $\angle KRA \cong \angle BGE$, $\angle TRG$ is a supplement of $\angle RGW$,
and $m\angle DWZ = 87^\circ$.



- a. What piece of information is contradictory?
- b. How do you know it is contradictory?
- c. Share your contradiction with your partner. Discuss any differences.

- d. Create four different pieces of information about this figure using $\overleftrightarrow{AV} \parallel \overleftrightarrow{BZ}$. One of the pieces of information should be contradictory.

1.

2.

3.

4.

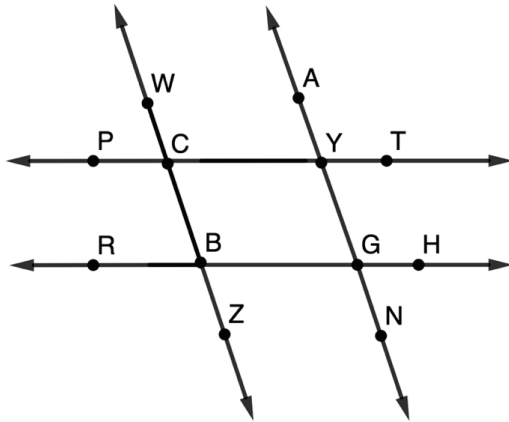
- e. Trade information with a partner and see if your partner can find the contradiction. Write their contradiction in the table.

Partner's Contradiction	Partner's Initials

2. Terrence was organizing a card sort for the proof, but his cards fell on the floor and are now out of order.

Given: $\overrightarrow{PT} \parallel \overrightarrow{RH}$ and
 $\angle CBG \cong \angle CYG$

Prove: $\overrightarrow{WZ} \parallel \overrightarrow{AN}$



Statement	Reason
A. $m\angle CBG = m\angle CYG$	K. Definition of Supplementary Angles
B. $\angle CBG$ and $\angle BGY$ are supplementary	L. Substitution Property of Equality
C. $\angle CBG \cong \angle CYG$	M. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.
D. $\angle CYG$ and $\angle BGY$ are supplementary	N. Definition of Congruence
E. $\overrightarrow{WZ} \parallel \overrightarrow{AN}$	P. Given
F. $m\angle CBG + m\angle BGY = 180^\circ$	R. If two parallel lines are intersected by a transversal, then the consecutive interior angles are supplementary.
G. $\overrightarrow{PT} \parallel \overrightarrow{RH}$	S. Definition of Supplementary Angles
H. $m\angle CYG + m\angle BGY = 180^\circ$	T. Given

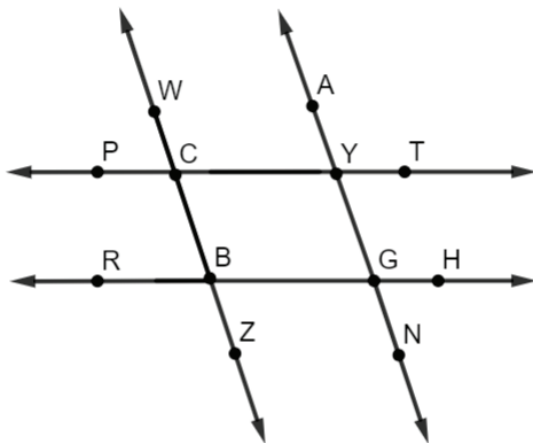
Work with your partner to correctly place the statements and reasons in the correct order by writing the letter.

Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.

3. Work with your partner to complete the proof in a different way.

Given: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ and $\angle CBG \cong \angle CYG$

Prove: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$



Statement	Reason
1. $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$	1. Given
2. $\angle CBG \cong \angle \underline{\hspace{2cm}}$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle CBG \cong \angle CYG$	3. Given
4.	4. Transitive Property of Congruence
5. $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$	5.

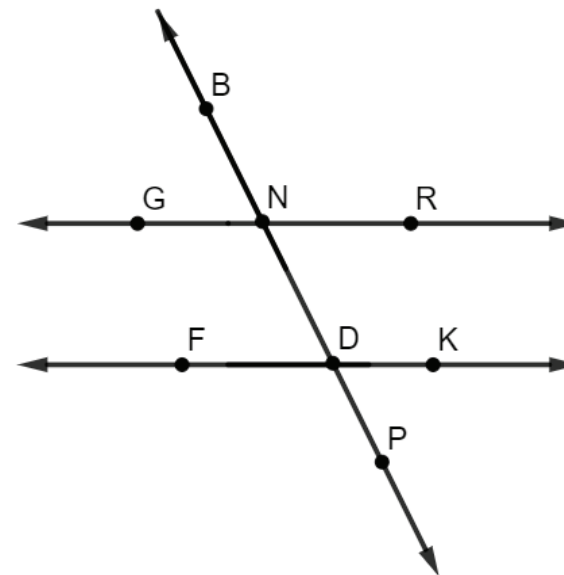


3.8.3 Guided Instruction: Proofs Involving Angle Relationship

1. Complete the flowchart proof.

Given: $\angle RND$ and $\angle FDP$ are supplementary.

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$



$\angle RND$ and $\angle FDP$ are supplementary. $\rightarrow$

$\angle FDP \cong \angle NDK$ $\rightarrow$

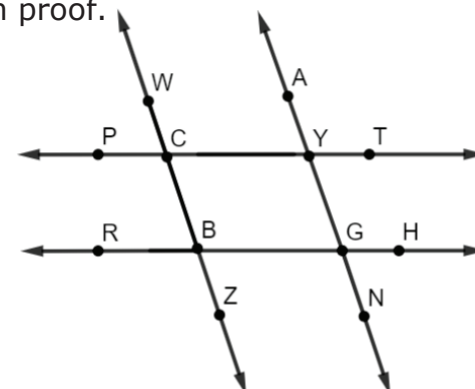
$\rightarrow$ $\angle RND$ and $\angle NDK$ are supplementary.

Substitution

2. Choose from the bank of statements and reasons to complete the paragraph proof.

Given: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ and $\angle PCW \cong \angle HGN$

Prove: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$



It is given that _____,

Therefore $\angle RBC \cong \angle PCW$ because if two parallel lines are intersected by a transversal then _____.

It is also given that _____. By the _____,

it can be written that $\angle RBC \cong \angle HGN$. Therefore, it can be concluded that $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$ because if two lines are intersected by a transversal so that _____ then the lines are parallel.

Statements and Reasons

- | | |
|--|--|
| <ul style="list-style-type: none"> • $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ • $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$ • $\angle PCW \cong \angle HGN$ • $\angle ZBG \cong \angle HGN$ • $\angle GND$ • $\angle FDP$ | <ul style="list-style-type: none"> • Definition of Congruence • Transitive Property of Congruence • Definition of Supplementary Angles • Corresponding angles are congruent. • Alternate exterior angles are congruent. • Alternate interior angles are congruent. |
|--|--|



3.8.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Using <i>substitution</i> as a reason in a proof.	  
Using the <i>transitive property of congruence</i> as a reason in a proof.	  
Using the <i>definition of congruence</i> as a reason in a proof.	  
Understanding what angles are congruent when two lines are parallel.	  
Understanding what angles are supplementary when two lines are parallel.	  

Unit 3, Lesson 9: Angle Relationship Proofs – Part 2



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

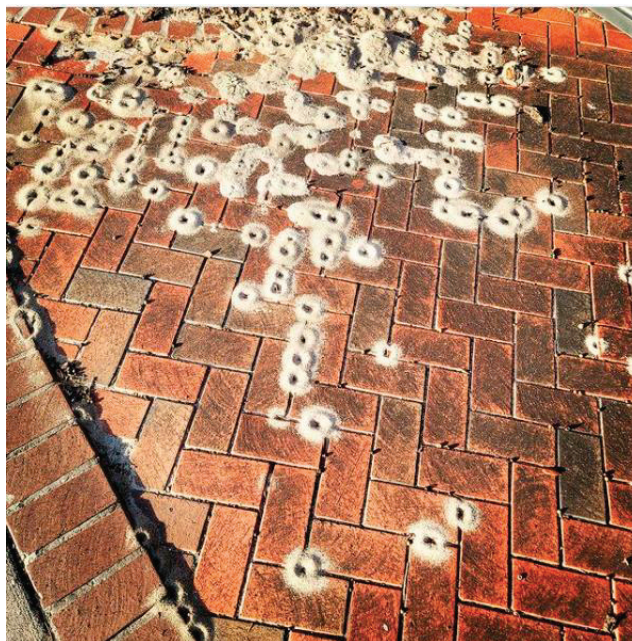
- ☐ I can prove relationships using parallel lines and their transversals.



3.9.1 Warm-Up: Notice and Wonder

A photo of ant hills taken in Gainesville, Florida is shown.

1. What do you notice?



2. What do you wonder?

3. Write a caption for the photo.



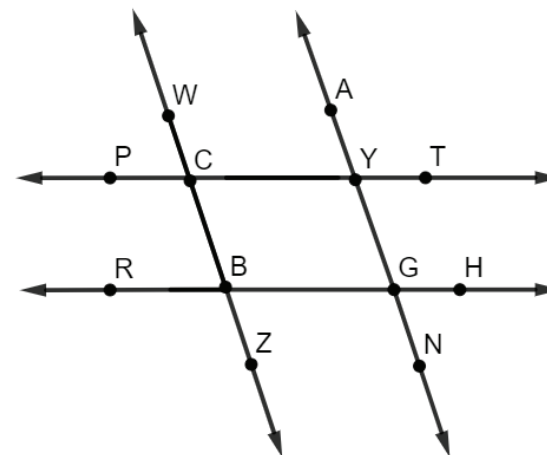
3.9.2 Activity: Card Sort

1. With your group, sort through the statements and reasons for each proof. Then, write each proof in the tables provided.

Proof A

Given: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$,
 $\angle WCY \cong \angle YGH$

Prove: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$

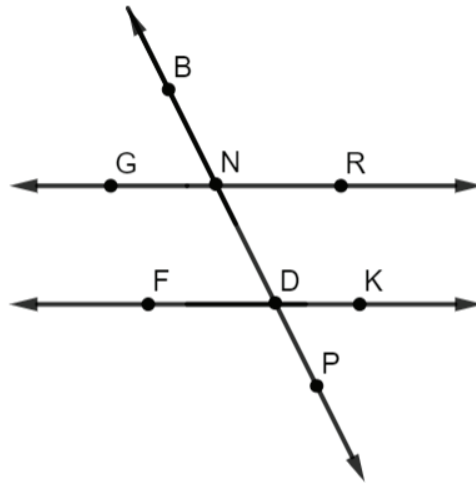


Statement	Reason

Proof B

Given: $\angle GND$ and $\angle PDK$ are supplementary.

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

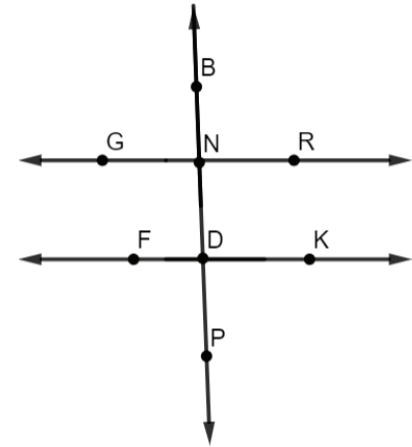


Statement	Reason

Proof C

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$, $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

Prove: $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

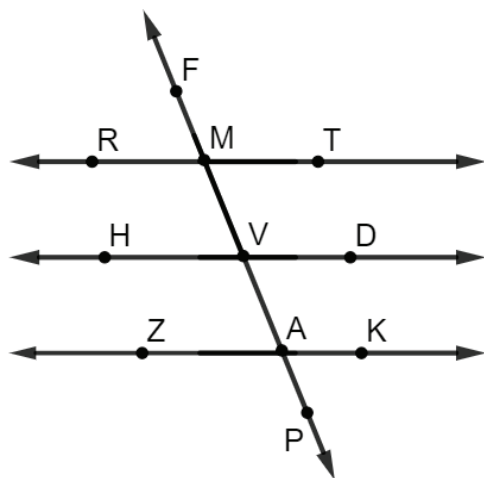


Statement	Reason

Proof D

Given: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$,
 $\angle TMV \cong \angle AVD$

Prove: $\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$

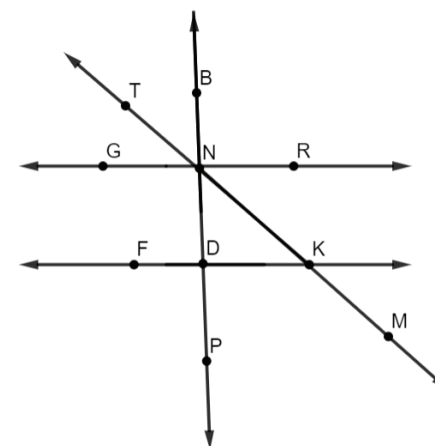


Statement	Reason

Proof E

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$ and
 $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

Prove: $\angle DKM$ and
 $\angle GNT$ are
supplementary.

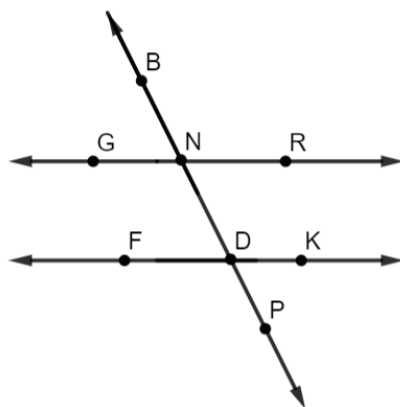


Statement	Reason

Proof F

Given: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

Prove: $\angle GND$ and $\angle PDK$ are supplementary.

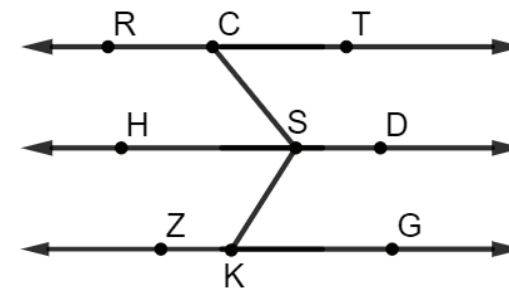


Statement	Reason

Proof G

Given: $\angle TCS \cong \angle HSC$,
 $\angle HSK \cong \angle SKG$

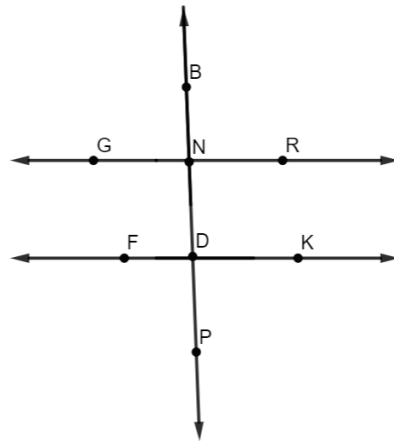
Prove: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$



Statement	Reason

Proof H

Given: $\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ and
 $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$
 Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$



Statement	Reason



3.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about what you've learned about angle relationship proofs. Include things that are easy to remember and common errors to watch out for.